

Exercise Set 5

Problem 1

A regression estimates the number of road accidents (acc) using daily time series data. The following explanatory variables are included: temperature ($temp$, measured on the Celsius scale) and a dummy variable indicating weekends ($weekend = 1$, if day of the week is Saturday or Sunday; $= 0$ otherwise). Temperature squared is also included in the model ($tempsq$). The model specification is:

$$\ln(acc_t) = \beta_1 + \beta_2 temp_t + \beta_3 tempsq_t + \beta_4 weekend_t + \varepsilon_t$$

The Stata output below shows results from an OLS estimation.

Source	SS	df	MS	Number of obs = 365		
Model	128.921938	3	42.9739794	F(3, 361) = 15.63		
Residual	992.318274	361	2.74880408	Prob > F = 0.0000		
Total	1121.24021	364	3.08033025	R-squared = 0.1150		
				Adj R-squared = 0.1076		
				Root MSE = 1.658		

ln_acc	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
temp	-.1006878	.0166068	-6.06	0.000	-.133346	-.0680295
tempsq	.0044969	.0007287	6.17	0.000	.0030639	.00593
weekend	-.5620591	.1924058	-2.92	0.004	-.9404361	-.1836821
_cons	7.190896	.1288277	55.82	0.000	6.937549	7.444243

1.1 Give the exact interpretation of the estimated coefficient on *weekend*.

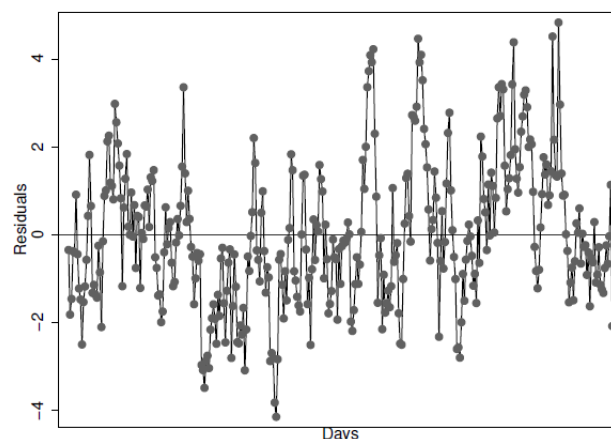
- The number of road accidents is, on average, associated with a decrease of $(\exp(-0.562) - 1) \times 100\% \approx 43\%$ on weekends, holding other factors constant.

1.2 Interpret the marginal effect of temperature for a given temperature of 10 degrees Celsius.

- Given a temperature of 10 degrees, the number of road accidents is associated with a decrease, on average, by 1% when temperature increases by 1 degree, ceteris paribus.

$$\frac{\partial \ln(acc_t)}{\partial temp_t} = \beta_2 + 2\beta_3 \cdot 10 = -0.101 + 2 \cdot 0.0045 \cdot 10 = -0.011$$

1.3 The graph below shows the residuals obtained from the regression. Interpret the pattern of the residuals.



- Residuals subsequent in time, i.e. e_t and e_{t-1} , tend to have both negative or positive values. Hence, $Cov(e_t, e_{t-1}) \neq 0$.
- The graph suggests that there is positive autocorrelation.

1.4 Carry out a Breusch-Godfrey test of autocorrelation. Test for a first-order autoregressive process at the 5% significance level. Give the null and alternative hypotheses, test statistic (including the auxiliary regression), degrees of freedom, rejection rule, and your test decision. *Note:* use the Stata output below, where the variable `L1.e` denotes the lagged residual e_{t-1} .

Source	SS	df	MS	Number of obs = 364		
Model	636.157566	4	159.039391	F(4, 359)	=	160.36
Residual	356.042705	359	.991762409	Prob > F	=	0.0000
Total	992.200271	363	2.73333408	R-squared	=	0.6412
				Adj R-squared	=	0.6372
				Root MSE	=	.99587

e	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
e						
L1.	.8033894	.0317211	25.33	0.000	.741007	.8657719
temp	.0034219	.0099779	0.34	0.732	-.0162006	.0230443
tempsq	-.0000395	.0004385	-0.09	0.928	-.0009019	.0008228
weekend	.1284414	.1157486	1.11	0.268	-.0991891	.3560719
_cons	-.0647561	.0774248	-0.84	0.404	-.2170193	.0875071

- Hypotheses: $H_0: \rho = 0$ vs. $H_1: \rho \neq 0$
- Test statistic: using R^2 from the auxiliary regression for $t = 2, \dots, T$

$$e_t = \alpha + \rho e_{t-1} + \mathbf{x}_t' \boldsymbol{\beta} + v_t,$$

the test statistic is:

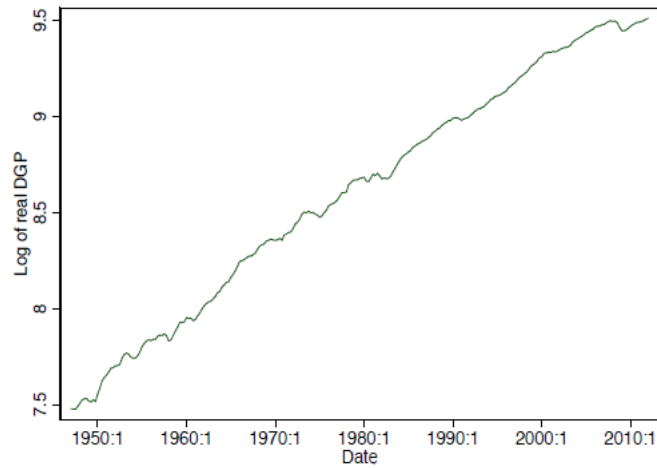
$$\chi^2 = (T-1)R^2 \sim \chi_1^2,$$

with 1 degree of freedom.

- Rejection rule: We reject H_0 if $\chi^2 > \chi_{1,0.05}^2 = 3.841$
- Calculation: $363 \cdot 0.64 = 232.96$
- Decision: We reject H_0 because $232.96 > 3.84$. There is evidence for autocorrelation.

Problem 3

3.1 The graph below shows the log of the United States real GDP over the period 1947 to 2012. Is the series stationary? Explain.



- The graph shows a linear trend in log of real GDP. Hence, it is nonstationary.

3.2 Show $E(\epsilon_t)$ for an AR(1) process. Make the following assumptions: (i) $\epsilon_t = \rho\epsilon_{t-1} + v_t$, (ii) $Var(v_t) = \sigma_v^2$, and (iii) Gauss-Markov assumptions are satisfied for v_t .

Expected Value:

$$\begin{aligned} E(\epsilon_t) &= E(\rho\epsilon_{t-1} + v_t) \\ &= \rho E(\epsilon_{t-1}) + E(v_t) \end{aligned}$$

$$E(\epsilon_t) = E(\epsilon_{t-1}):$$

$$E(\epsilon_t) = \rho E(\epsilon_t) + E(v_t)$$

$$E(\epsilon_t) = \frac{1}{(1-\rho)} \underbrace{E(v_t)}_{=0} = 0$$

Problem 4

4.1 Explain the autocorrelation function (ACF) and the partial autocorrelation function (PACF) of an autoregressive process of order p .

- ACF is infinite in extent (it tails off).
- PACF is (close to) zero for lags larger than p .

4.2 Explain the autocorrelation function (ACF) and the partial autocorrelation function (PACF) of a moving average process of order q .

- ACF is (close to) zero for lags larger than q .
- PACF is infinite in extent (it tails off).

4.3 Suppose you want to investigate the time series properties of the inflation rate in the US over the years from 1960 to 2010 by means of an augmented Dickey-Fuller (ADF) test. (i) State the null hypothesis and the alternative hypothesis. (ii) Specify the ADF test regression when two lags are included. (iii) Carry out the ADF test using the Stata output below ($\alpha = 0.1$). (iv) Give your decision.

Augmented Dickey-Fuller test for unit root		Number of obs = 201		
		----- Interpolated Dickey-Fuller -----		
	Test	1% Critical	5% Critical	10% Critical
	Statistic	Value	Value	Value
z(t)	-3.145	-3.476	-2.883	-2.573

- Hypotheses: H_0 : unit root vs. H_1 : not H_0
- ADF test regression:

$$\Delta Y_t = \delta + \pi Y_{t-1} + c_1 \Delta Y_{t-1} + c_2 \Delta Y_{t-2} + \varepsilon_t$$

- Carrying out the test: We reject the null hypothesis of a unit root at the 10% significance level because $-3.145 < -2.573$.
- Decision: we conclude that there is no (strong) evidence for a unit root.

4.4 Compute a measure of inflation persistence using the estimation results from the AR(3) model shown below. What is your conclusion?

ARIMA regression

Sample: 1960q2 - 2011q1

Number of obs = 204

Wald chi2(3) = 283.64

Log likelihood = -463.6308

Prob > chi2 = 0.0000

	infl	Coef.	OPG Std. Err.	z	P> z	[95% Conf. Interval]	
infl	_cons	3.781947	1.020659	3.71	0.000	1.781492	5.782402

ARMA

	ar						
	L1.	.2925791	.0579182	5.05	0.000	.1790614	.4060968
	L2.	.2277573	.0613714	3.71	0.000	.1074715	.3480431
	L3.	.2969597	.0534582	5.55	0.000	.1921836	.4017358
	/sigma	2.342933	.0538576	43.50	0.000	2.237374	2.448492

- An appropriate measure of persistence is the sum of coefficients in the autoregressive process:

$$\sum_{j=1}^p \theta_j.$$

Note that $\sum_{j=1}^p \theta_j = 1$ corresponds to a unit root and infinite persistence.

- Here, we have $.293 + .228 + .297 = 0.818$, suggesting that persistence is quite high.
- Other approaches are possible.

4.5 Suppose a Ljung-Box test for 12 lags is carried out for an AR(3) model. The empirical value of the test statistic is 16.67. Give the null hypothesis and the alternative hypothesis, number of degrees of freedom, rejection rule ($\alpha = 0.01$), and test decision.

- Hypotheses: H_0 : error term is white noise vs. H_1 : not H_0 .
- The test has generally $K - p - q$ degrees of freedom, where K is the number of lags being tested. p and q denote the order of the AR and MA processes, respectively. Here, we have $12 - 3 = 9$ degrees of freedom.
- We reject H_0 at the 1% significance level if the empirical values of the test statistic is larger than $\chi^2_{9,0.01} = 21.67$.
- Decision: We do not reject H_0 . We conclude that there is no (strong) evidence against the assumption of a white noise error.

References

Wooldridge, J. (2013). *Introductory Econometrics. A modern approach*.

Verbeek, M. (2012). *A Guide to Modern Econometrics*. John Wiley & Sons, Chichester.